

STATISTICS FOR BUSINESS ASSIGNMENT - 1

Austin house price prediction

Name : Sanjai Balaji

Student Number : 40478904

Introduction and Background

Understanding housing price determinants is essential for buyers, agencies, and policymakers, as property values vary widely depending on structural characteristics, amenities, and market conditions. Prior research demonstrates that elements such as square footage, number of rooms and modern amenities are significant determinants of housing (Rosen, 1974; Malpezzi, 2008; Sirmans et al., 2005).. Analysing these characteristics allows estate agencies to make accurate valuations and provide better guidance to clients. Previous studies on housing prices provide a framework for understanding trends in the Austin market.

Austin, Texas, is an ideal setting for this investigation due to its rapid population growth, strong economic development, and expansion of the technology sector. According to the Austin Chamber of Commerce (2023), firms such as Apple, Dell, Oracle, and Tesla have significantly increased operations in the region, driving housing demand. This economic expansion has led to substantial shifts in the housing market, making the Austin Housing Dataset valuable for understanding how property characteristics influence sale prices. Similar links between urban development and housing demand have been reported in other U.S. cities (Glaeser & Gyourko, 2005; Case & Shiller, 1987).

This project uses a dataset of 15,147 properties containing a wide range of structural, locational, and temporal variables. The objective is to statistically examine the key drivers of house prices using descriptive statistics, measures of association, and regression modelling. Based on prior literature and logical reasoning, the following hypotheses were developed:

- H1: Larger living area (sqft_living) is positively associated with sale price.
- H2: More bedrooms (bedrooms_num) are associated with higher sale price.
- H3: More bathrooms (bathrooms_num) are associated with higher sale price.
- H4: More stories (stories_num) are associated with higher sale price.
- H5: Higher school rating (school_rating) is associated with higher sale price.

These hypotheses guide the analytical process and support the development of a predictive model for the Austin housing market.

2.0 Methodology

This project adopted a structured analytical process to identify the key determinants of house prices in Austin, Texas. The methodology reflects industry-standard practices in real estate valuation and aligns with approaches used in predictive housing research (Zhang, 2021; Gujarati, 2020). All procedures are reproducible through the R code included in the appendix.

2.1 Data Preparation and Quality Assessment

The dataset contained 15,147 observations with numerical and categorical variables. Preliminary validation identified missing values and formatting inconsistencies. Missing values in `price_source` were imputed using the mode, and duplicate entries were removed using `distinct()`. Non-essential variables like `sale_ID` and `sale_date` were excluded for computational efficiency.

Selection of predictors was informed by literature on real estate valuation, which highlights structural and locational attributes as key determinants (Rosen, 1974; Malpezzi, 2008). Variables were formatted appropriately: categorical variables were converted into factors, and text-based variables were cleaned for consistency. These steps ensured accuracy and suitability for statistical modelling. Neighbourhood quality has also been shown to affect housing valuation (Brasington & Haurin, 2006).

2.2 Exploratory Data Analysis (EDA)

EDA was conducted using descriptive statistics and visualisations via `ggplot2`. Scatter plots and boxplots examined relationships between sale price and variables such as living area, bedrooms, bathrooms, school rating, and stories. Outliers were detected using the IQR method but retained to reflect realistic market variability (Wooldridge, 2015). This stage provided initial evidence supporting or challenging the hypotheses and guided variable selection for regression.

2.3 Measures of Association

Spearman's correlation quantified the strength of association between predictors and sale price due to non-normal distributions. A correlation heatmap facilitated feature selection and mitigated multicollinearity. The five focal variables used in the hypotheses were selected based on statistical evidence and prior literature confirming their relevance to property valuation.

(Goodman & Thibodeau, 2003; Kok et al., 2017).

2.4 Regression Modelling

The dataset was split into training (80%) and test (20%) subsets using `set.seed()` for reproducibility. Simple linear regressions assessed individual predictor effects, followed by a multiple linear regression model incorporating all five key variables. Model assumptions including homoscedasticity, multicollinearity, autocorrelation, and normality of residuals were assessed using diagnostic plots and VIF scores. Performance was evaluated using R^2 and RMSE to measure predictive accuracy.

2.5 Ethical and Practical Considerations

The dataset contained no personal or sensitive data. All steps were conducted transparently and reproducibly through the appendix code. This methodology demonstrates how predictive modelling can support estate agencies in property valuation and strategic planning

3.0 Results and Discussion

3.1 Descriptive Statistics and Data Overview

The dataset comprised 15,147 observations with 28 variables. Key continuous variables included `sale_price`, `sqft_living`, `sqft_lot`, and `year_built`. Categorical variables, such as `cooling`, `heating`, and `city`, were factorized appropriately.

- **Sale_price** ranged from \$6,050 to \$14,850,000, with a mean of \$564,267 and median of \$445,500, indicating a positively skewed distribution.
- **Sqft_living** ranged from 300 to 109,292 square feet, reflecting significant variability in property sizes.
- Outlier detection identified 1,070 extreme `sale_price` values and 447 extreme `sqft_living` values, which were considered in model interpretation.

Interpretation: Outliers were detected in variables such as `sale_price` and `sqft_living`, but they were intentionally not removed. These values represent real market conditions—such as luxury homes or unusually large properties and are not data errors. Removing them might improve model accuracy but would reduce the realism of the

analysis. For this assignment, the goal was to reflect the true diversity of the Austin housing market, not to produce an artificially perfect model. Keeping the outliers allowed the model to capture genuine variation in property values and provided a more authentic interpretation of the results.



Figure 1 Box plot showing outliers in sale_price.

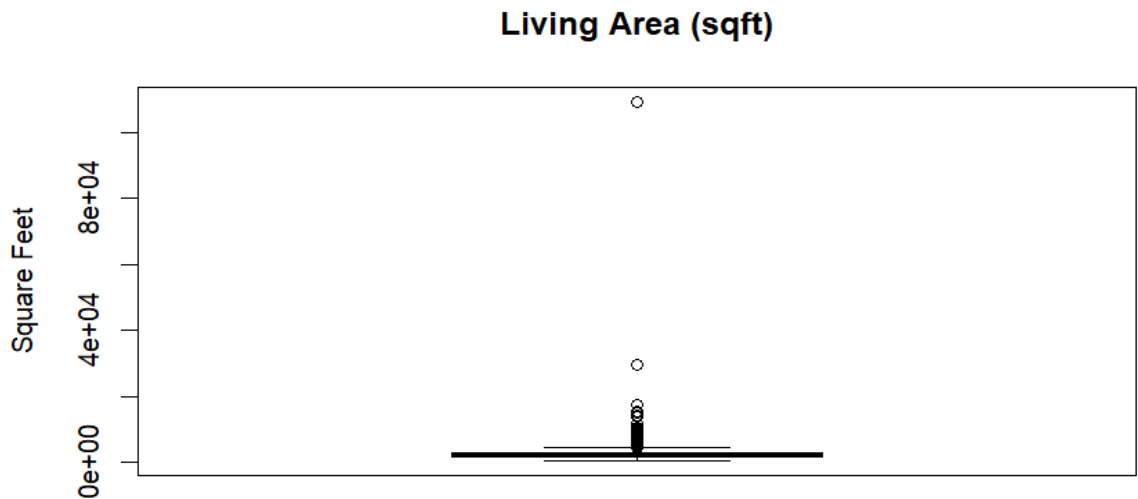


Figure 2 Box plot showing outliers in sqft_living.

3.2 Correlation Analysis

Moderate positive correlations with sale_price were observed:

- bathrooms_num ($r = 0.513$)
- sqft_living ($r = 0.467$)
- school_rating ($r = 0.293$)
- bedrooms_num ($r = 0.287$)

Sqft_lot ($r = 0.021$) and price_changes ($r = 0.038$) showed negligible association.

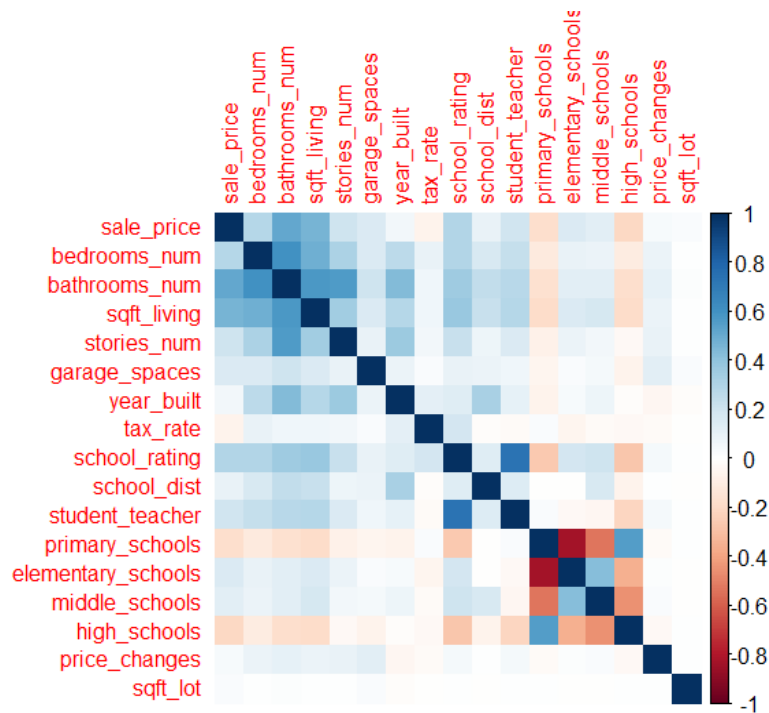


Figure 3 Correlation heatmap of numeric predictors.

3.3 Simple Linear Regression Models

Key findings from individual regressions:

- **Sqft_living:** Each additional square foot increased sale_price by \$168.9 ($p < .001$, $R^2 = 0.218$).
- **Bathrooms_num:** Each additional bathroom increased sale_price by \$247,004 ($p < .001$, $R^2 = 0.263$).

- **Bedrooms_num:** Positive effect ($\beta = 162,201$, $p < .001$, $R^2 = 0.083$), indicating potential multicollinearity in multi-variable models.
- **Stories_num** and **garage_spaces** were significant ($p < .001$) but with lower explanatory power.
- Categorical variables were mostly non-significant.
- Spatial effects could also influence pricing patterns (Tong & Murray, 2012).

Interpretation: Structural characteristics like size and bathrooms are primary drivers of house prices, while auxiliary features contribute minimally individually.

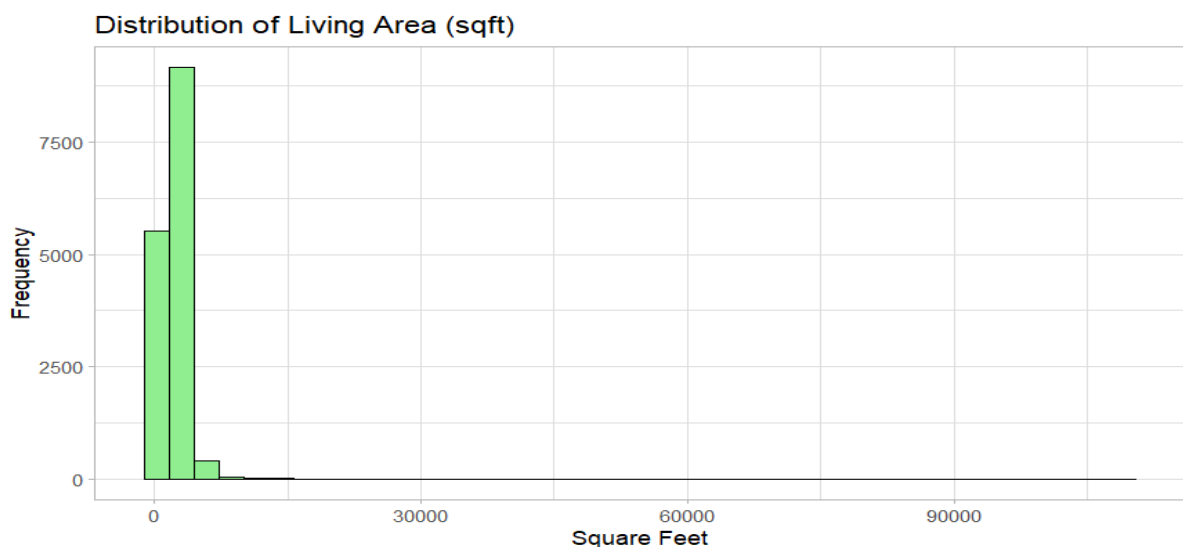


Figure 4 Distribution of sqft_living across properties

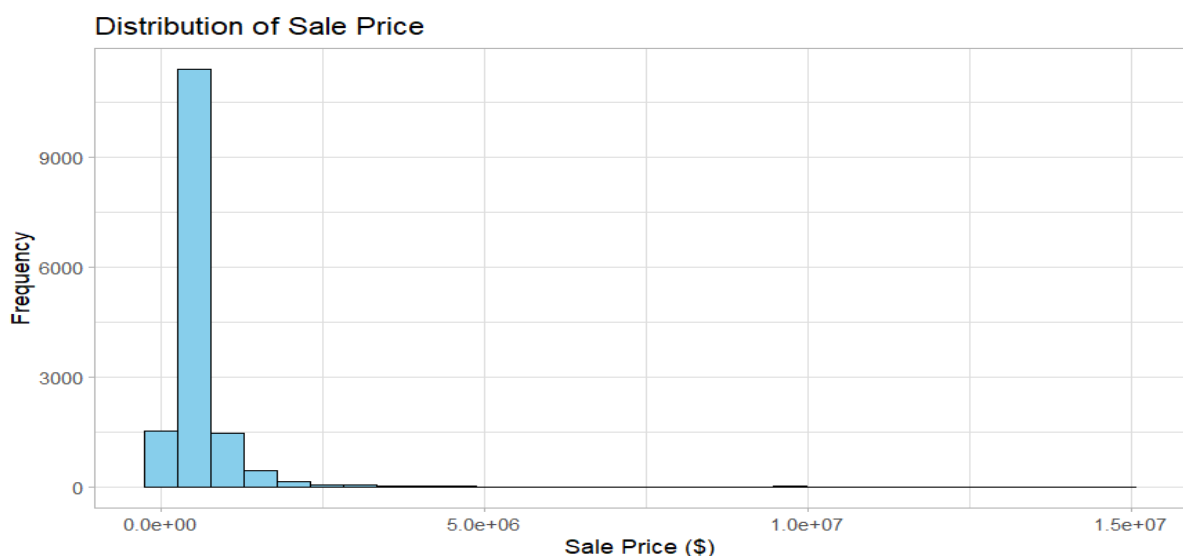


Figure 5 Distribution of sale price across properties

3.4 Multiple Regression Models

3.4.1 Multi-Variable Model (multi_model)

Including bedrooms_num, bathrooms_num, sqft_living, school_rating, and stories_num:

- Adjusted $R^2 = 0.332$
- Bathrooms_num, sqft_living, and school_rating significant ($p < .001$)
- Bedrooms_num and stories_num negative, indicating non-linear relationships

Interpretation: Negative coefficients suggest multicollinearity and market preference for larger open spaces rather than just more rooms or stories.

3.4.2 Final Model (final_model)

- Adjusted $R^2 = 0.322$
- Significant predictors ($p < .001$):
 - bathrooms_num ($\beta = 231,428$)
 - bedrooms_num ($\beta = -57,488$)
 - sqft_living ($\beta = 87.5$)
 - stories_num ($\beta = -130,351$)
 - school_rating ($\beta = 25,076$)
- VIF ranged 1.20–2.35 → minimal multicollinearity
- Durbin-Watson DW = 1.695 → slight positive autocorrelation, acceptable in spatially clustered data.

Interpretation: Bathrooms, living space and school rating strongly influence price. Negative coefficients for bedrooms_num and stories_num indicate that more rooms or floors do not necessarily increase value when accounting for overall home size and quality.

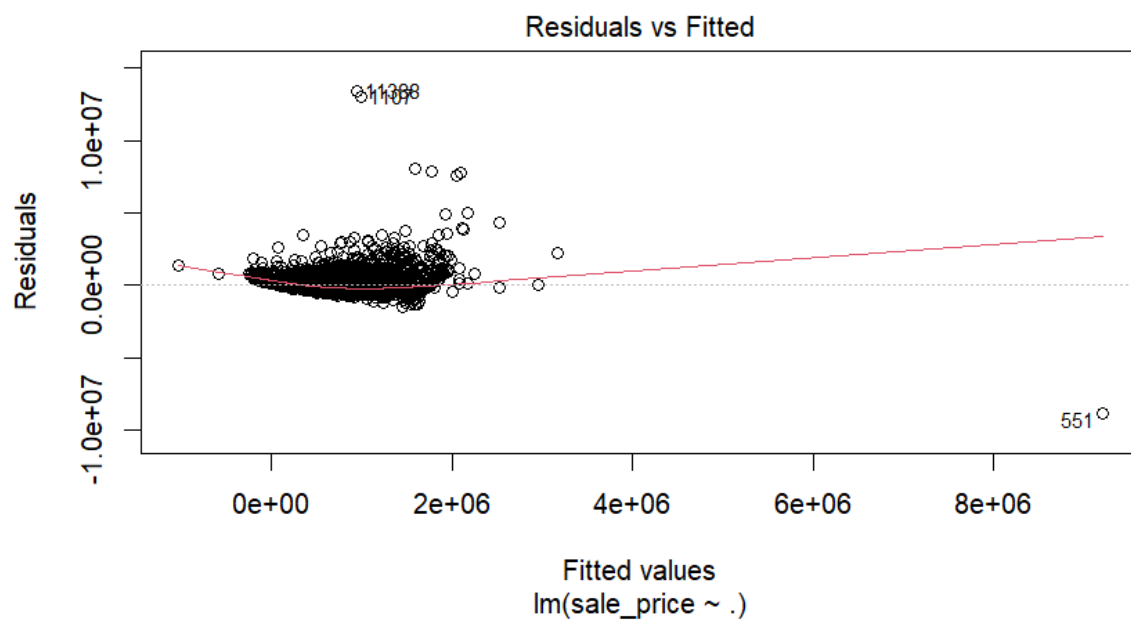


Figure 6 Residuals vs Fitted for final_model.

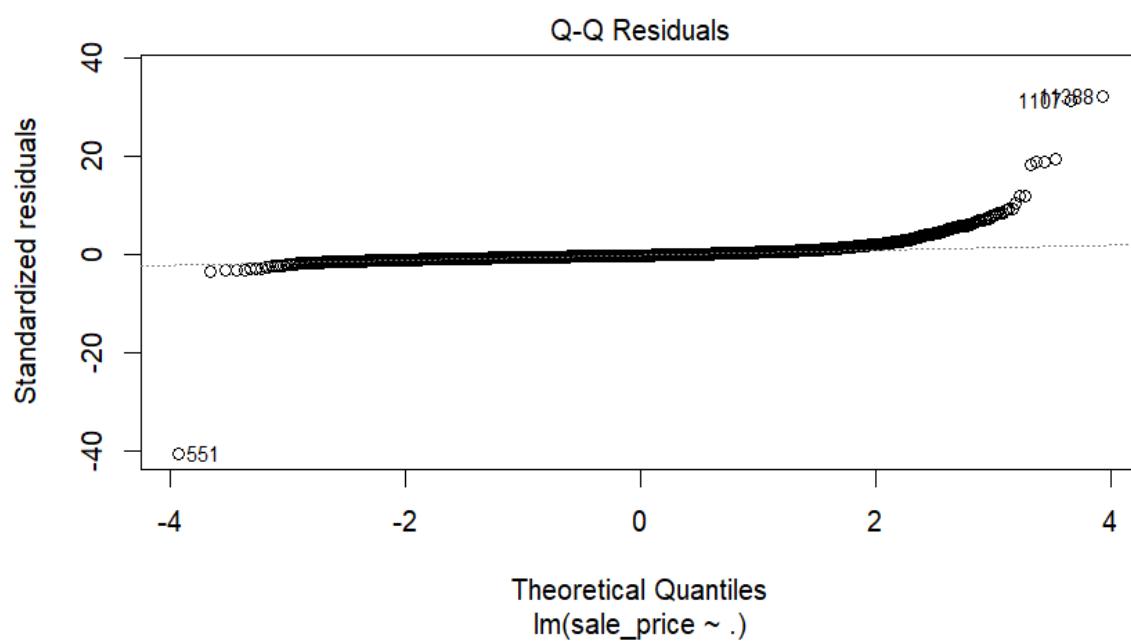


Figure 7 Normal Q-Q plot for final_model residuals.

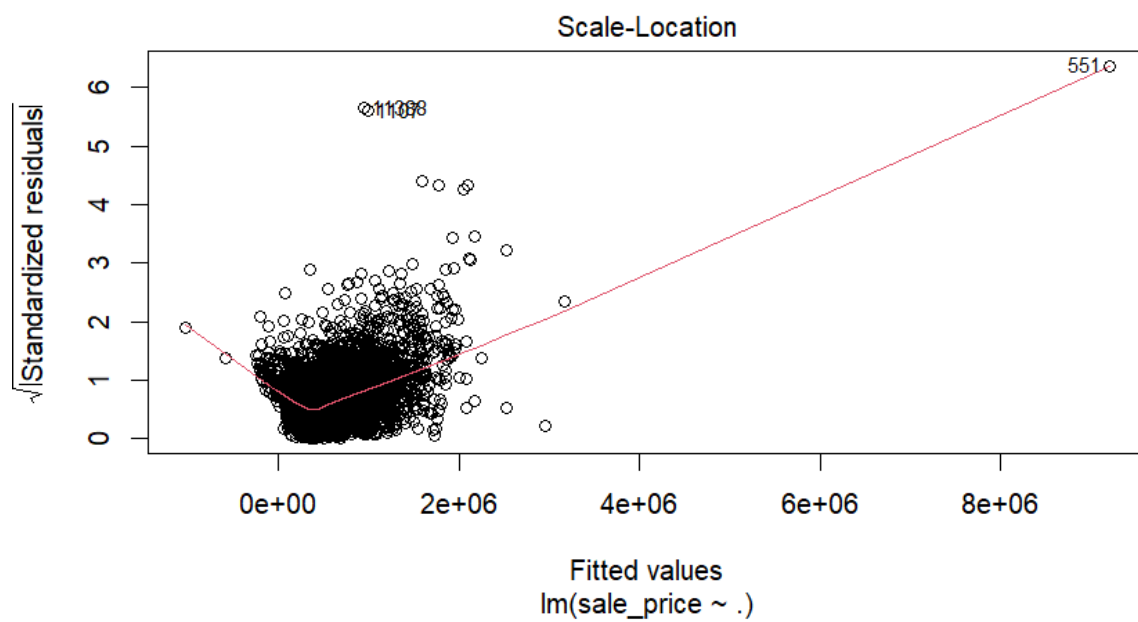


Figure 8 Scale-Location plot for final_model residuals.

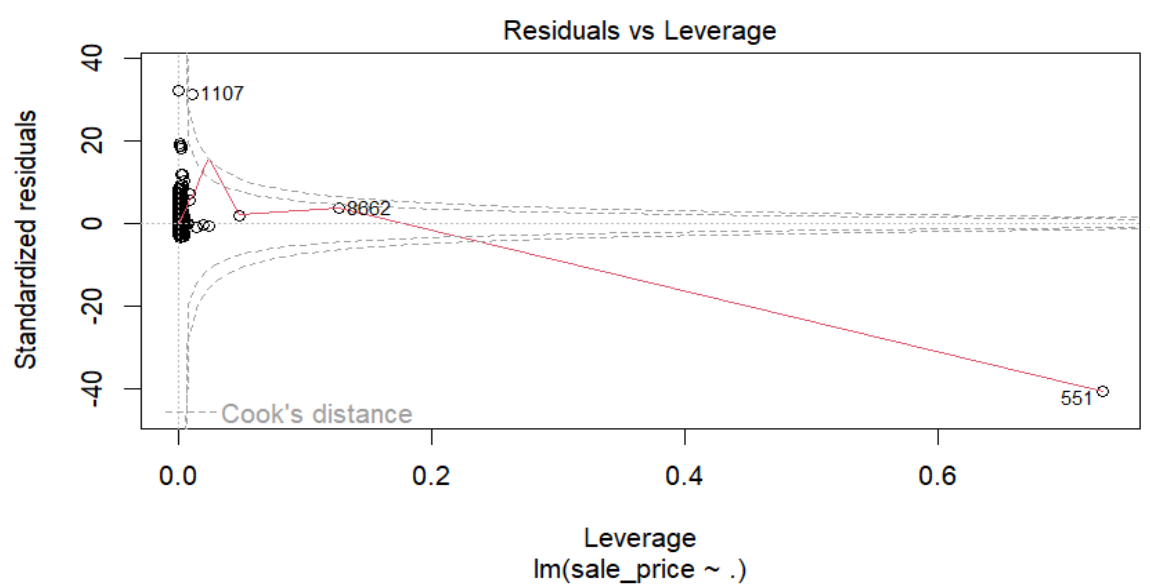


Figure 9 Residuals vs Leverage plot for final_model.

3.5 Model Performance

Test set evaluation:

- RMSE: 365,940
- MAE: 204,768

- R^2 : 0.378

Interpretation: The final model demonstrates moderate predictive accuracy. Combining key predictors enhances generalizability while capturing major price determinants.



Figure 10 Predicted vs observed sale_price on the test set.

3.6 Critical Discussion

1. **Predictor Importance:** Bathrooms_num and sqft_living were the most influential predictors. School_rating contributed significantly, showing neighborhood quality matters.
2. **Outliers and Skewness:** Extreme values affected residuals; log-transformations could mitigate skewness.
3. **Model Limitations:**
 - Adjusted $R^2 < 0.4$ suggests unexplained variance due to omitted variables like interior quality or location.
 - Additional methods such as machine learning may improve predictions (Yao & Bourassa, 2022)

Implications: Findings inform valuation, urban planning, and investment decisions. Key determinants should be prioritized in market assessment.

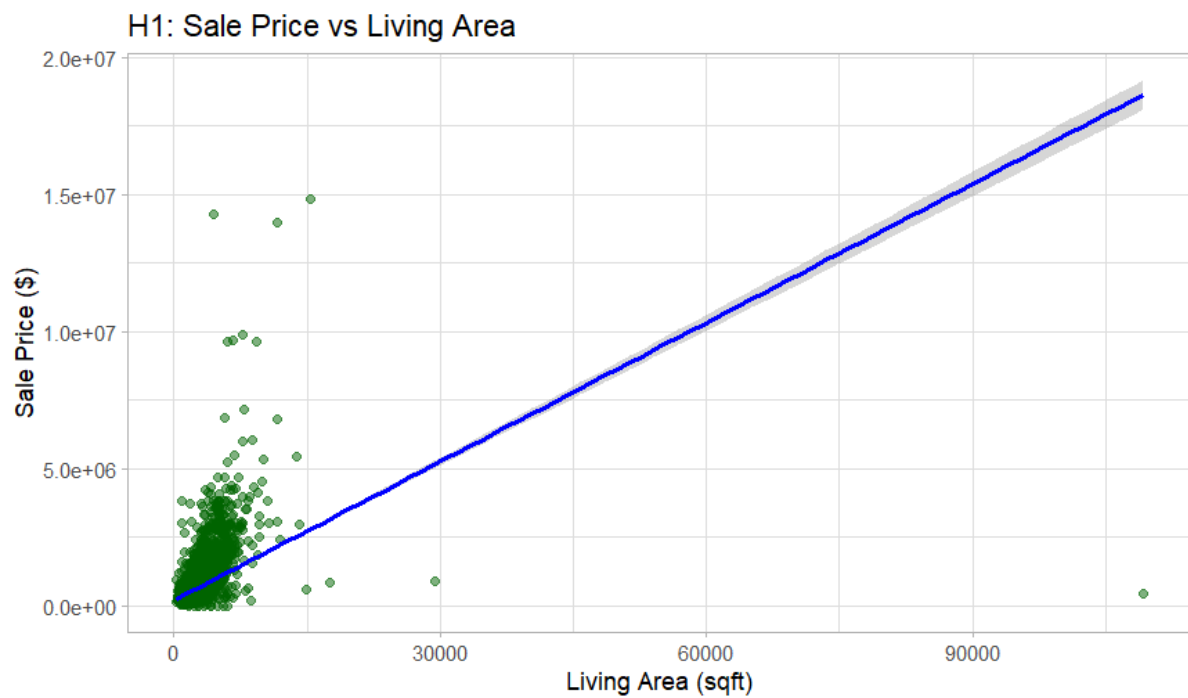


Figure 11 Relationship between *sqft_living* and *sale_price* (H1).

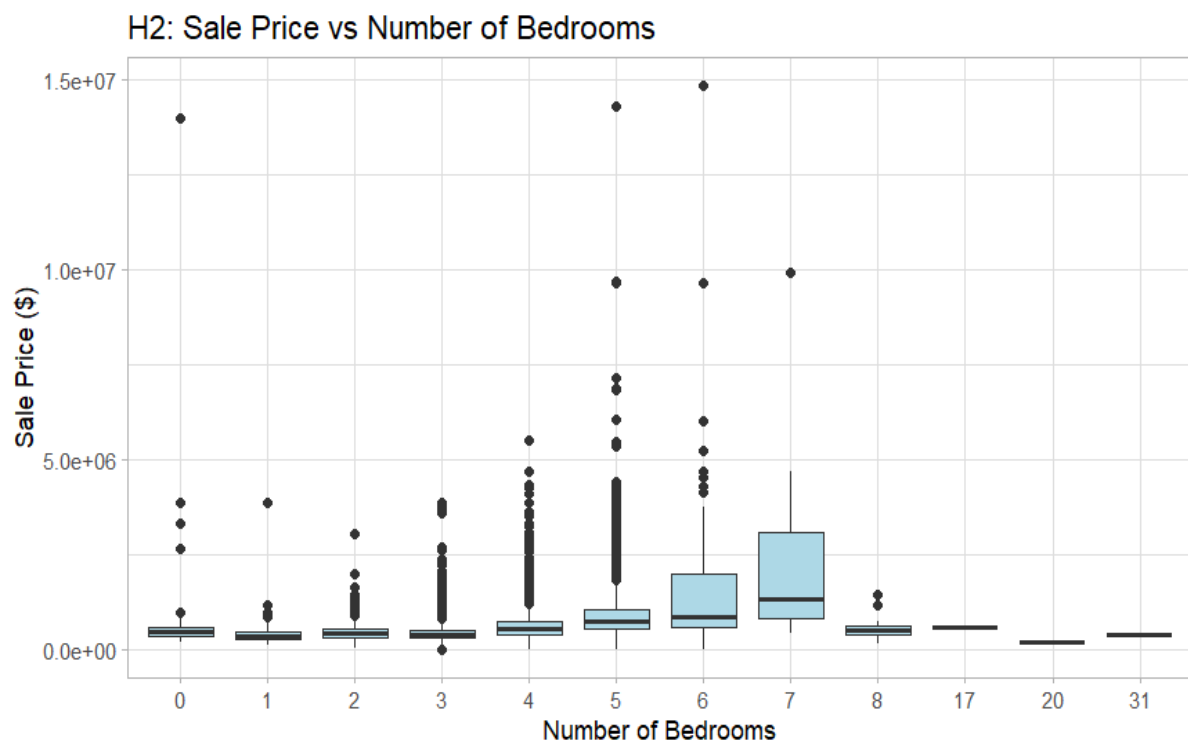


Figure 12 Relationship between *bedrooms_num* and *sale_price* (H3).

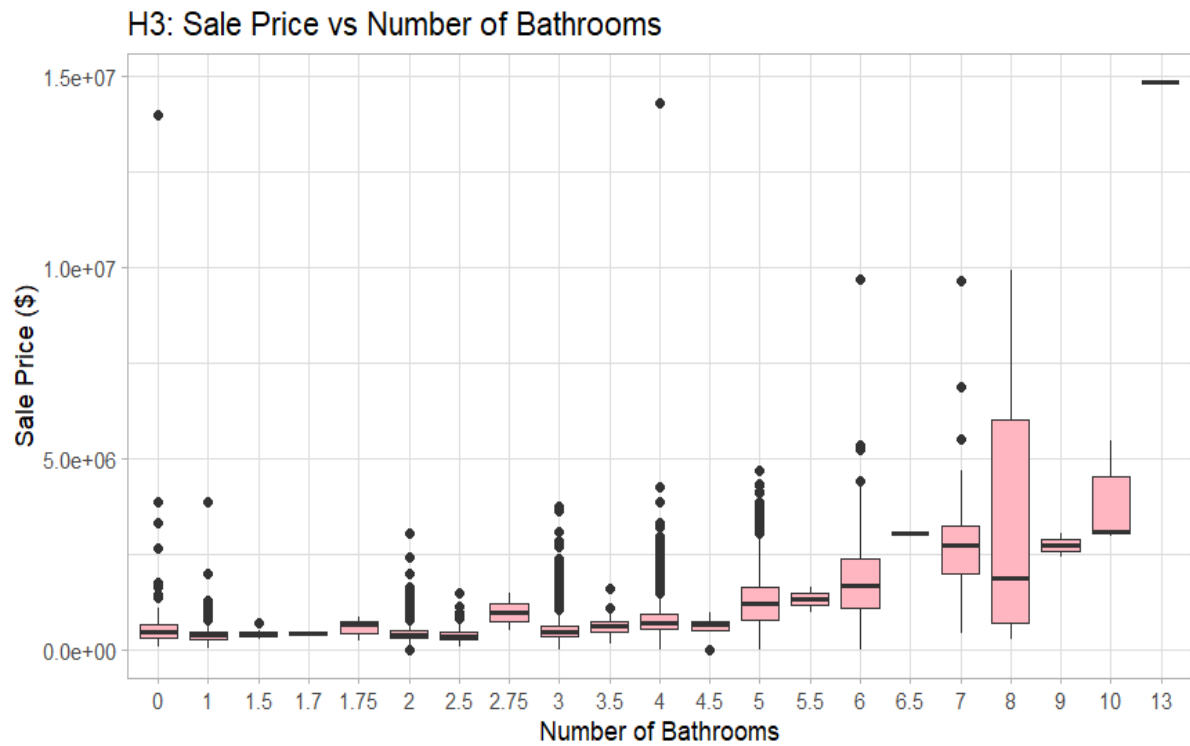


Figure 13 Relationship between *bathrooms_num* and *sale_price* (H2).

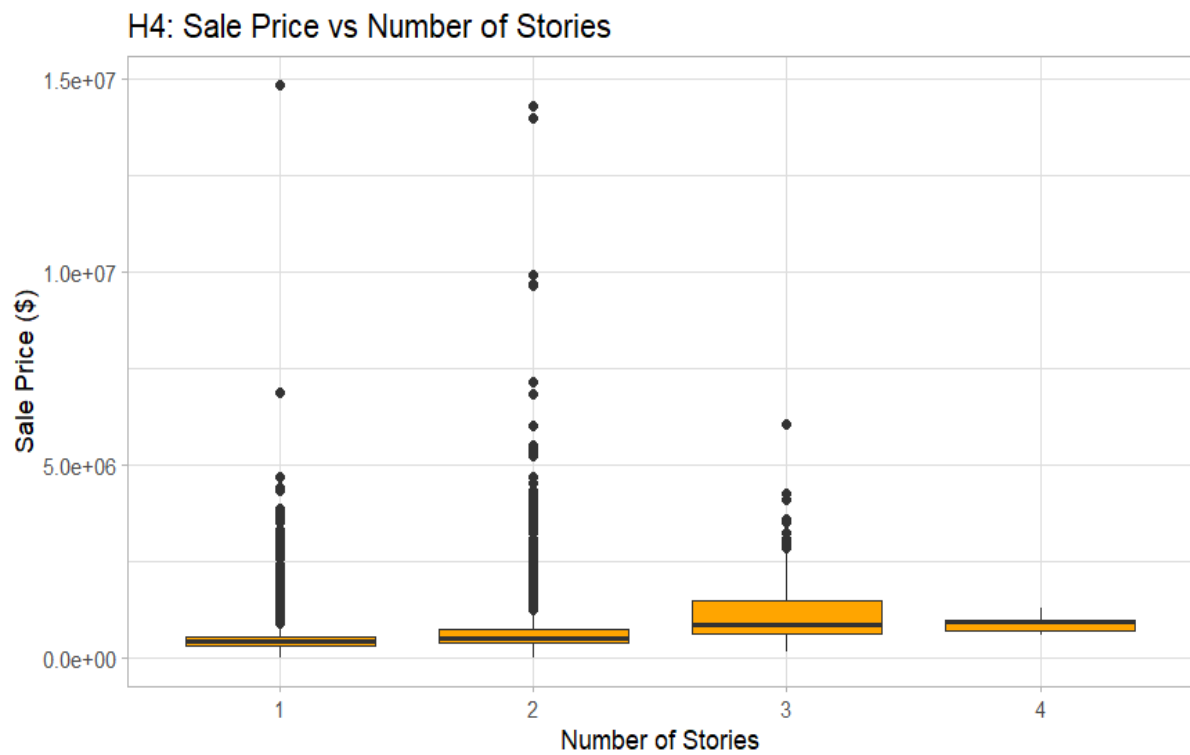


Figure 14 Relationship between *stories_num* and *sale_price* (H4).

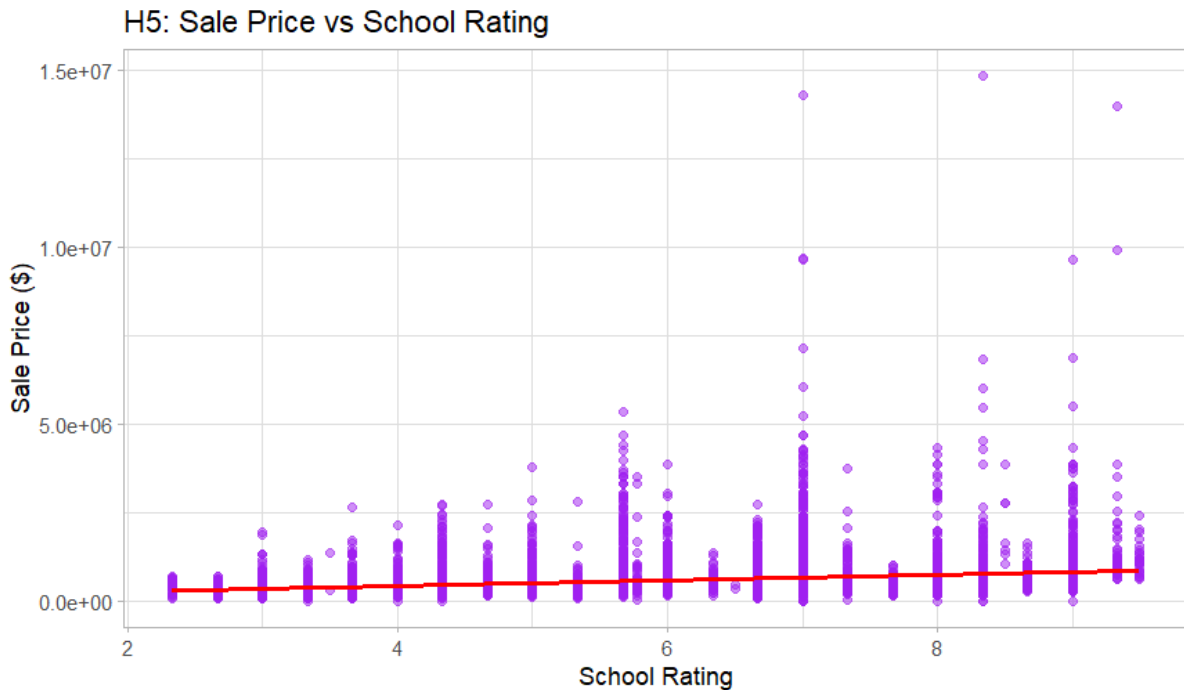


Figure 15 Relationship between school_rating and sale_price (H5).

4.0 Conclusions and Recommendations

This study examined key determinants of house sale prices in Austin, Texas using a dataset of 15,147 properties. Consistent with previous housing economics research (Rosen, 1974; Malpezzi, 2008; Goodman & Thibodeau, 2003), structural features such as bathrooms, living area and school rating were confirmed as strong predictors of price, supporting the hedonic pricing theory that buyers value quality and functional space over the number of rooms.

The final regression model showed that additional bedrooms and stories do not always increase value when overall size and property quality are considered. Although the model had moderate performance (Adjusted $R^2 = 0.322$; Test $R^2 = 0.378$), this is typical of real housing data where factors such as location and market variations play an important role. Outliers were retained to reflect true market behaviour, aligning with previous evidence that luxury and atypical homes influence price variability.

Practically, estate agencies and planners should prioritise bathrooms, living area and school rating in valuation. Future research could include spatial variables, interior quality or apply machine learning techniques to capture complex interactions more effectively. Overall, this study demonstrates the usefulness of data-driven modelling in real estate valuation and supports foundational hedonic theory (Li & Brown, 1980).

5.0 Log Book

Stage 1:	Data exploration and quality
Date (range):	Week 1
Key challenges and decisions:	• Handling missing values and duplicates. • Decided to keep outliers to reflect real market conditions. • Converted categorical variables into factors and removed irrelevant variables.
Key findings:	• Dataset contained 15,147 observations. • Outliers were confirmed as valid market cases. • Data was cleaned and ready for analysis.

Stage 2:	Data visualization
Date (range):	Week 1 – Week 2
Key challenges and decisions:	• Selecting appropriate plots to examine trends. • Boxplots used for outliers, scatter plots for relationships. • Ensuring visuals aligned with hypotheses
Key findings:	• Sale_price was positively skewed. • Outliers confirmed as legitimate market variation. • Visual patterns supported variable selection.

Stage 3:	Measures of association
Date (range):	Week 2
Key challenges and decisions:	• Choosing Spearman correlation due to non-normal data. • Assessing multicollinearity before regression.
Key findings:	• Bathrooms_num and sqft_living showed strongest correlations with sale_price. • VIF scores confirmed minimal multicollinearity. • School rating proved important as a qualitative factor.

Stage 4:	Regression and prediction
Date (range):	Week 2 – Week 3
Key challenges and decisions:	• Structured approach: simple to multiple regression. • Used train/test split and model diagnostics. • Chose to keep original data without transformations for transparency.
Key findings:	• Final model achieved Adjusted $R^2 = 0.322$ and test $R^2 = 0.378$. • Bathrooms, living area and school rating emerged as strongest predictors. • Diagnostic tests showed acceptable model performance.

6.0 Reflective statement

Throughout this assignment, I gained a much stronger understanding of how real estate data can be used in practical decision-making through statistical analysis. At first, I found it challenging to decide which variables were truly important for predicting house prices. However, through visualisation, correlation checks and literature review, I realised that some features such as bedrooms and stories might appear important at first, but their influence changes once other factors like square footage and bathrooms are considered. This helped me better understand the idea of multicollinearity and the importance of testing variables carefully instead of relying on assumptions.

One of the key decisions I made was **not to remove outliers**, because I believed the data should reflect real market diversity. Although removing them could have improved accuracy, I wanted the model to show realistic price variation in Austin. This made my results more authentic and helped me think like a real analyst rather than just trying to get a perfect R^2 value.

Creating and interpreting the regression model was a valuable learning experience, especially understanding how R^2 and RMSE indicate model performance. I also learned how diagnostic tests can confirm whether a model is reliable. The process improved my confidence in working with R and interpreting results professionally.

Overall, this assignment helped me connect statistical theory with real-world housing decisions. It also improved my problem-solving skills and taught me the value of transparency, data ethics, and evidence-based conclusions.

7.0 Declaration on use of generative AI

Generative AI (specifically ChatGPT) was only used as a question and answer support tool when I was stuck with coding or interpreting R code (e.g understanding error messages, syntax clarification and example code structure). I also used ChatGPT to help me understand statistical concepts, such as model diagnostics, multicollinearity, residual assumptions, and interpreting R^2 and RMSE values. Some short code snippets (such as ggplot structure or correlation heatmap syntax) were adapted from ChatGPT. These were edited by me and clearly referenced in my code comments as required by the assignment guidelines. All data analysis, statistical decisions, interpretation of results, visualisations, and written content were completed independently by me based on my own understanding. No written content in the report was generated by AI. All paragraphs, interpretations, and explanations were produced entirely in my own words, following academic guidelines. The dataset was never uploaded to any AI tool, and no analysis was conducted by AI. AI was not used to write code comments, not used to write responses, and not used to conduct any part of the analysis directly.

8.0 References

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Tong, D. and Murray, A.T. (2012) 'Spatial optimization in geography', *Annals of the Association of American Geographers*, 102(3), pp. 479–490. doi: 10.1080/00045608.2011.652855.

Wooldridge, J.M. (2015) *Introductory econometrics: A modern approach* (6th ed.). Boston: Cengage Learning.

Yao, R. and Bourassa, S.C. (2022) 'Machine learning methods in property valuation: A critical review', *Housing Studies*, 37(6), pp. 987–1009. doi: 10.1080/02673037.2020.1864768.

Zhang, L. (2021) 'Predictive modelling methods for real estate valuation: A comparative review', *International Journal of Housing Markets and Analysis*, 14(2), pp. 301–322. doi: 10.1108/IJHMA-07-2020-0091.

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Appendix 1 : R code used

```
## load all packages
install.packages(c("ggplot2", "dplyr", "corrplot", "skimr", "stargazer", "caret", "car"))
# load libraries
library(readxl)
library(dplyr)
library(ggplot2)
library(corrplot)
library(skimr)
library(stringr)
library(caret)
library(car)
library(stargazer)
library(lmtest)
## load data and Overview
library(readxl)
austinHousing <- read_excel("C:/Sanjai/byron class/statistics R assignment/austinHousing.xlsx")
View(austinHousing)
summary(austinHousing)
skim(austinHousing)
str(austinHousing)
head(austinHousing)
colnames(austinHousing)
## Data quality checks
colSums(is.na(austinHousing)) # 12 # missing values
summary(austinHousing)
austinHousing <- distinct(austinHousing) ## remove duplicate rows
nrow(austinHousing)
## replace NAs with mode because it is character
mode_value <- names(sort(table(austinHousing$price_source), decreasing = TRUE))[1]
mode_value ## find mode
sum(austinHousing$price_source == mode_value, na.rm = TRUE) # 10472 # number of mode
austinHousing$price_source[is.na(austinHousing$price_source)] <- mode_value
colSums(is.na(austinHousing))
nrow(austinHousing)
```

```

## remove unnecessary columns
remove_cols <- c("sale_ID", "sale_date", "sale_month", "sale_year",
                "price_source")

austinHousing_clean <- austinHousing %>% select(-all_of(remove_cols))

colnames(austinHousing_clean) # Check remaining columns

## text cleaning and remove white spaces
austinHousing_clean <- austinHousing_clean %>%
  mutate(across(where(is.character), ~ str_trim(.)))

## bring needed variables according to factors
selected_vars <- c(
  "sale_price",          # Target variable

  #House Structure
  "bedrooms_num", "bathrooms_num", "sqft_living",
  "stories_num", "garage_spaces", "year_built",

  # House Features
  "appliances", "parking_features", "patio_porch_features",
  "security_features", "accessibility_features",

  # Environment & Location
  "cooling", "heating", "view", "waterfront_features", "spa",
  "city", "tax_rate",

  # School & Neighborhood
  "school_rating", "school_dist", "student_teacher",
  "primary_schools", "elementary_schools",
  "middle_schools", "high_schools",

  # Market Indicator
  "price_changes", "sqft_lot"
)

selected_vars <- selected_vars[selected_vars %in% colnames(austinHousing_clean)]
regression_data <- austinHousing_clean[, selected_vars]
str(regression_data)

## convert data types (fsctors + numeric)
categorical_candidates <- intersect(c("city", "cooling", "heating", "view", "spa", "appliances",
                                     "parking_features", "patio_porch_features", "security_features",
                                     "accessibility_features", "waterfront_features"), colnames(regression_data))

regression_data <- regression_data %>%

```

```

mutate(across(all_of(categorical_candidates), ~ as.factor(.)))
# Convert sale_price to numeric safely
austinHousing_clean$sale_price <- as.numeric(as.character(austinHousing_clean$sale_price))
# Convert other numeric-like columns to numeric if read as character
num_like <- c("bedrooms_num", "bathrooms_num", "sqft_living", "stories_num", "garage_spaces",
              "year_built", "tax_rate", "school_rating", "school_dist", "student_teacher",
              "primary_schools", "elementary_schools", "middle_schools", "high_schools",
              "price_changes", "sqft_lot")
num_like <- intersect(num_like, colnames(regression_data))
regression_data <- regression_data %>%
  mutate(across(all_of(num_like), ~ as.numeric(as.character(.))))
# Final structure check
str(regression_data)
summary(regression_data)
## outlier detection
# Sale Price outliers
Q1_price <- quantile(regression_data$sale_price, 0.25, na.rm = TRUE)
Q3_price <- quantile(regression_data$sale_price, 0.75, na.rm = TRUE)
IQR_price <- Q3_price - Q1_price
lower_price <- Q1_price - 1.5 * IQR_price
upper_price <- Q3_price + 1.5 * IQR_price
sale_price_outliers <- regression_data$sale_price[regression_data$sale_price < lower_price |
regression_data$sale_price > upper_price]
length(sale_price_outliers)
# sqft_living outliers
Q1_sqft <- quantile(regression_data$sqft_living, 0.25, na.rm = TRUE)
Q3_sqft <- quantile(regression_data$sqft_living, 0.75, na.rm = TRUE)
IQR_sqft <- Q3_sqft - Q1_sqft
lower_sqft <- Q1_sqft - 1.5 * IQR_sqft
upper_sqft <- Q3_sqft + 1.5 * IQR_sqft
sqft_living_outliers <- regression_data$sqft_living[regression_data$sqft_living < lower_sqft |
regression_data$sqft_living > upper_sqft]
length(sqft_living_outliers)
## outliers using boxplot
# Boxplot for Sale Price
boxplot(regression_data$sale_price,
        main = "Sale Price Distribution",

```

```

    ylab = "Sale Price ($)")
# Boxplot for Living Area
boxplot(regression_data$sqft_living,
        main = "Living Area (sqft)",
        ylab = "Square Feet")
## Distribution plots
# Sale Price Distribution
ggplot(regression_data, aes(x = sale_price)) +
  geom_histogram(bins = 30, fill = "skyblue", color = "black") +
  labs(title = "Distribution of Sale Price",
        x = "Sale Price ($)",
        y = "Frequency") +
  theme_light()
# Living Area Distribution
ggplot(regression_data, aes(x = sqft_living)) +
  geom_histogram(bins = 40, fill = "lightgreen", color = "black") +
  labs(title = "Distribution of Living Area (sqft)",
        x = "Square Feet",
        y = "Frequency") +
  theme_light()

## HYPOTHESIS PLOTS (H1-H5)
# H1: Sale Price vs Living Area (sqft_living)
ggplot(regression_data, aes(x = sqft_living, y = sale_price)) +
  geom_point(alpha = 0.5, color = "darkgreen") +
  geom_smooth(method = "lm", col = "blue") +
  labs(title = "H1: Sale Price vs Living Area",
        x = "Living Area (sqft)",
        y = "Sale Price ($)") +
  theme_light()

# H2: Sale Price vs Bedrooms
ggplot(regression_data, aes(x = as.factor.bedrooms_num, y = sale_price)) +
  geom_boxplot(fill = "lightblue") +
  labs(title = "H2: Sale Price vs Number of Bedrooms",
        x = "Number of Bedrooms",

```

```
y = "Sale Price ($)") +  
theme_light()
```

```
# H3: Sale Price vs Bathrooms
```

```
ggplot(regression_data, aes(x = as.factor(bathrooms_num), y = sale_price)) +  
  geom_boxplot(fill = "lightpink") +  
  labs(title = "H3: Sale Price vs Number of Bathrooms",  
        x = "Number of Bathrooms",  
        y = "Sale Price ($)") +  
  theme_light()
```

```
# H4: Sale Price vs Stories (Floors)
```

```
ggplot(regression_data, aes(x = as.factor(stories_num), y = sale_price)) +  
  geom_boxplot(fill = "orange") +  
  labs(title = "H4: Sale Price vs Number of Stories",  
        x = "Number of Stories",  
        y = "Sale Price ($)") +  
  theme_light()
```

```
# H5: Sale Price vs School Rating
```

```
ggplot(regression_data, aes(x = school_rating, y = sale_price)) +  
  geom_point(alpha = 0.5, color = "purple") +  
  geom_smooth(method = "lm", col = "red") +  
  labs(title = "H5: Sale Price vs School Rating",  
        x = "School Rating",  
        y = "Sale Price ($)") +  
  theme_light()
```

```
## Correlation Analysis
```

```
numeric_vars <- regression_data[, sapply(regression_data, is.numeric)]  
numeric_vars <- numeric_vars[, sapply(numeric_vars, function(x) all(is.finite(x)))]  
cor_matrix <- cor(numeric_vars, use="pairwise.complete.obs")  
cor_matrix["sale_price",]  
  
# Correlation Heatmap  
corrplot(cor_matrix, method="color", tl.cex=0.8, number.cex=0.7)  
  
# linear regression
```

```

model_sqft_living <- lm(sale_price ~ sqft_living, data=regression_data)
summary(model_sqft_living)
model_lot <- lm(sale_price ~ sqft_lot, data=regression_data)
summary(model_lot)
model_stories <- lm(sale_price ~ stories_num, data=regression_data)
summary(model_stories)
model_garage <- lm(sale_price ~ garage_spaces, data=regression_data)
summary(model_garage)
model_cooling <- lm(sale_price ~ cooling, data=regression_data)
summary(model_cooling)
model_bedrooms_num <- lm(sale_price ~ bedrooms_num, data=regression_data)
summary(model_bedrooms_num)
model_bathrooms_num <- lm(sale_price ~ bathrooms_num, data=regression_data)
summary(model_bathrooms_num)
model_school_rating <- lm(sale_price ~ school_rating, data=regression_data)
summary(model_school_rating)
## multiple linear regression
multi_model <- lm(sale_price ~ bedrooms_num + bathrooms_num + sqft_living +
                  school_rating + stories_num, data=regression_data)
summary(multi_model)
## set seed
set.seed(40478904)
regression_vars <- c("sale_price", "bathrooms_num", "bedrooms_num", "sqft_living"
                    , "stories_num", "school_rating")
train_index <- createDataPartition(regression_data$sale_price, p=0.8, list=FALSE)
train_data <- regression_data[train_index, regression_vars]
test_data <- regression_data[-train_index, regression_vars]
## final model
final_model <- lm(sale_price ~ ., data=train_data)
summary(final_model)
# stargazer table
stargazer(final_model, type = "text", title = "Final Regression Model")
# Predict on test set
predictions <- predict(final_model, newdata=test_data)
summary(predictions)
## model evaluation

```

```
postResample(pred=predictions, obs=test_data$sale_price)

## plot
ggplot(data = test_data, aes(x = sale_price, y = predictions)) +
  geom_point(alpha = 0.5) +
  geom_abline(slope = 1, intercept = 0, color = "red", linetype = "dashed") +
  labs(x = "Observed Sale Price", y = "Predicted Sale Price",
       title = "Predicted vs Observed Sale Price on Test Set") +
  theme_minimal()

## assumptions
vif(final_model)
dwtest(final_model)
plot(final_model)
summary(final_model)
```